Sheet 5 - (SVD)

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Question 1-Find the SVD for the following matrices:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} \boldsymbol{3} & \boldsymbol{-1} \\ \boldsymbol{1} & \boldsymbol{3} \\ \boldsymbol{1} & \boldsymbol{1} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

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Question :2

Compute $A^{\mathrm{T}}A$ and AA^{T}, and their eigenvalues and unit eigenvectors, for

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Multiply the three matrices $U\Sigma V^{\mathrm{T}}$ to recover A.

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Question :3

1. Show that for any orthogonal matrix Q, all singular values = 1
2. Prove: rank(A) = # non-zero singular values
3. Show: $\|A\|_2 = \sigma_1$ (largest singular value)

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Question :4

Problems 1–7 bring out the underlying ideas of the SVD.

1. Suppose $\boldsymbol{u_1}, \dots, \boldsymbol{u_n}$ and $\boldsymbol{v_1}, \dots, \boldsymbol{v_n}$ are orthonormal bases for $\mathbb{R}^n$. Construct the matrix A that transforms each $\boldsymbol{v_j}$ into $\boldsymbol{u_j}$ to give $\boldsymbol{Av_1} = \boldsymbol{u_1}, \dots, \boldsymbol{Av_n} = \boldsymbol{u_n}$.
2. Construct the matrix with rank 1 that has $Av = 12u$ for $v = \frac{1}{2}(1,1,1,1)$ and $u = \frac{1}{3}(2,2,1)$. Its only singular value is $\sigma_1 =$ ___.
3. Find $\boldsymbol{U\Sigma V^T}$ if A has orthogonal columns $w_1, \dots, w_n$ of lengths $\sigma_1, \dots, \sigma_n$.
4. Explain how $\boldsymbol{U\Sigma V^T}$ expresses A as a sum of r rank-1 matrices in equation (3):

$$A = \sigma_1 u_1 v_1^T + \cdots + \sigma_r u_r v_r^T.$$

5. Suppose A is a 2by 2symmetric matrix with unit eigenvectors u_1 and u_2. If its eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = -2$, what are U, Σ, and V^T?
6. Suppose A is invertible (with $\sigma_1 > \sigma_2 > 0$). Change A by as small a matrix as possible to produce a singular matrix A_0. Hint: U and V do not change:

   Find A_0 from

$$A = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{\mathrm{T}}$$

7. (a) If A changes to $4A$, what is the change in the SVD?

   (b) What is the SVD for A^T and for A^{-1}?